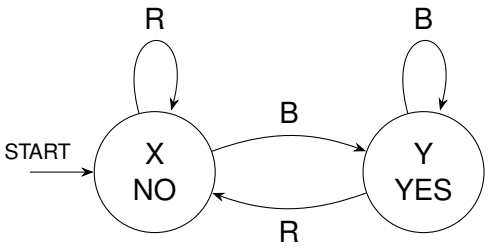
Read cards from left to right. Start fresh for each new row. Mark one fixed starting state START. Each state has a fixed YES or NO and exactly one fixed R arrow and one fixed B arrow. Follow the arrow for each card, even if it stays put. Hide used cards: the marker's state is the whole memory. Stop anywhere and read YES or NO.

This example machine says YES when the last card is blue.



Input: R B B

$X \xrightarrow{R} X \xrightarrow{B} Y \xrightarrow{B} Y$

Output: YES

Problem 1: Build a machine that says YES exactly when two red cards next to each other have appeared anywhere. It must still say YES after more cards arrive. How few states can it use?

R R B B R B R B R R B

Problem 2: Build a machine that says YES exactly when the row has an even number of red cards and an even number of blue cards. Zero counts as even. How few states can it use?

Row	Machine says
No cards	
R	
R B	
R R B B	
B B R	
B R B R	

Problem 3: Build a machine that says YES exactly when the red and blue counts are equal. Try to defeat a partner's machine with a card row. Could any machine with a fixed number of states work for rows of every length? Explain your answer.

Row	Machine says